

SS $= Q[\Phi^{\rm P}, \Phi^{\rm P}]$: closed form in 3D; modes $\sim A_\ell/\Delta r^2$ near r_0

First-order residual $\Phi^{\rm res} = \Phi_1 - \Phi^{\rm P}$ on the real line ($\ell' \le \ell'_{\max}$)

RR $= Q[\Phi^{\rm res}, \Phi^{\rm res}]$: smooth; real line throughout

SS on the contour: complex-r angular quadrature of the 3D source at each contour node; smooth integrand, no subtraction

Taylor data at $\mathbf{x}_0$ from mode sums at r_0 of the *same* truncated residual: $T_k\Phi^{\rm res} = \Phi^{\rm res}(\mathbf{x}_0) + \Delta x^i \partial_i \Phi^{\rm res}(\mathbf{x}_0) + \dots$

subtract

SR Part 1: $2Q[\Phi^{\rm P}, T_k\Phi^{\rm res}]$ closed form; carries the $\delta(r - r_0)$ terms, e.g. from $\Phi^{\rm res}(\mathbf{x}_0)\,\partial_z^2\rho^{-1}$; complex-$r$ quadrature ($\delta \to \delta_{\rm ext}$ poles)

SR Part 2: $2Q[\Phi^{\rm P}, \Phi^{\rm res} - T_k\Phi^{\rm res}]$ formed on the real line from real fields (no shadow); integrable, no δ terms if $k \ge d - 2$ ($d = 2$); solved on the real line (or Cauchy-continued onto the contour)

Contour solves inside the worldtube, real line outside: $\tilde{s}$ = SS, and $\tilde{s}$ = SR Part 1 (plus Part 2 if continued). Every contour source is linear in one continued quantity, so there is no shadow cross-term. Take Re at a_R.

Real-line solve of SR Part 2 (if not continued)

Φ_2 outside the worldtube = $\Phi_2^{\rm SS} + \Phi_2^{\rm SR} + \Phi_2^{\rm RR}$ (linear superposition of the three solves)